Case Study 2

AKSTA Statistical Computing

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2026-04-02

The .Rmd **and** *.html (or .pdf) should be uploaded in TUWEL by the deadline. Refrain from using explanatory comments in the R code chunks but write them as text instead. Points will be deducted if the submitted file is not in a decent form.*

DISCLAIMER: In case students did not contribute equally, include a disclaimer stating what each student's contribution was.

Table 1: AI usage table.

AI usage	**a)**	**b)**	**c)**	**d)**	**e)**	**f)**	**g)**	**h)**
Correcting faulty code	no	no	no	no	no	no	no	no
Improving existing code	no	no	no	no	no	no	no	no
Generating new code	no	no	no	no	no	no	no	no
Layout and formatting	no	no	no	no	no	no	no	no
Text correction	no	no	no	no	little	little	little	no

The CIA World Factbook provides intelligence on various aspects of 266 world entities, including history, people, government, economy, energy, geography, environment, communications, transportation, military, terrorism, and transnational issues. This case study involves analyzing world data from 2020, focusing on:

- **Education Expenditure (% of GDP)**
- **Youth Unemployment Rate (15-24 years)**
- **Net Migration Rate** (difference between the number of people entering and leaving a country per 1,000 persons)

The data was sourced from the CIA World Factbook Archives. You are required to use `dplyr` for data manipulation, while any package can be used for importing data.

Tasks:

```r
packages <- c("tidyverse", "countrycode")

install_if_missing <- packages[!(packages %in% installed.packages()[,"Package"])]
if(length(install_if_missing)) install.packages(install_if_missing)


library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.4.3
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
## Warning: package 'tibble' was built under R version 4.4.3

## Warning: package 'tidyr' was built under R version 4.4.3

## Warning: package 'readr' was built under R version 4.4.3

## Warning: package 'purrr' was built under R version 4.4.3

## Warning: package 'dplyr' was built under R version 4.4.3

## Warning: package 'stringr' was built under R version 4.4.3

## Warning: package 'forcats' was built under R version 4.4.3

## Warning: package 'lubridate' was built under R version 4.4.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 4.4.3
```

```r
library(countrycode)
```

```
## Warning: package 'countrycode' was built under R version 4.4.3
```

a. Data Import and Cleaning

Load the following datasets from TUWEL and ensure that missing values are handled correctly and column names are clear. Each dataset should ultimately contain only two columns: **country** and the respective variable. Note that some data sets also contain information on the year when the value was last updated.

- `rawdata_369.txt` which contains the (estimated) public expenditure on education as a percent of GDP. *Pay attention! The delimiter is 2 or more white spaces (one space would not work as it would separate country names which contain a space); you have to skip the first two lines.*
- `rawdata_373.csv` which contains the (estimated) youth unemployment rate (15-24) per country
- `rawdata_347.txt` which contains (estimated) net migration rate per country.

```r
ed_exp_headers <- c("index", "ed_country_name", "ed_expenditure", "exp_year")
mig_rate_headers <- c("index", "mig_country_name", "mig_rate", "mig_year")
custom_countries <- c("Kosovo" = "Kosovo",
                      "Saint Martin" = "Saint Martin",
                      "Virgin Islands" = "Virgin Islands",
                      "West Bank" = "West Bank",
                      "Gaza Strip" = "Gaza Strip")

ed_exp <- read_lines("rawdata_369.txt", skip = 2) %>%
  str_replace_all(" {2,}", "\t") %>%
  paste(collapse = "\n") %>%
  read_tsv(col_names = ed_exp_headers, na = c("", "NA", "N/A")) %>%
```

```r
  mutate(country_name = countrycode(ed_country_name, origin = 'country.name', destination = 'country.nar
  select("country_name", "ed_expenditure")
```

```
## Rows: 170 Columns: 4
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (1): ed_country_name
## dbl (3): index, ed_expenditure, exp_year
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
mig_rate <- read_lines("rawdata_347.txt", skip = 2) %>%
  str_replace_all(" {2,}", "\t") %>%
  paste(collapse = "\n") %>%
  read_tsv(col_names = mig_rate_headers, na = c("", "NA", "N/A")) %>%
  mutate(country_name = countrycode(mig_country_name, origin = 'country.name', destination = 'country.na
  select("country_name", "mig_rate")
```

```
## Rows: 227 Columns: 4
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (2): mig_country_name, mig_year
## dbl (2): index, mig_rate
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
unemployment <- read_csv("rawdata_373.csv", trim_ws = TRUE) %>%
  mutate(country_name = countrycode(country_name, origin = 'country.name', destination = 'country.name'
```

```
## Rows: 181 Columns: 2
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): country_name
## dbl (1): youth_unempl_rate
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

b. Merging Raw Data

Merge the datasets using `dplyr` on a unique key and retain the union of all observations.

- What key are you using for merging?

For merging the datasets, we use country_name as the key. But since the country names can be different across the datasets, we first standardize them using the countrycode() function so that we have consistent naming across the datasets. Special cases like Kosovo, West Bank and Gaza Strip are handled with the custom_match argument, as countrycode() cannot match them and would otherwise return NA.

- Return the dimensions of the merged dataset.

The dimensions of the merged dataset are: 227 rows and 4 columns.

```r
ed_exp_final <- ed_exp %>%
  group_by(country_name) %>%
  summarise(ed_expenditure = mean(ed_expenditure, na.rm = TRUE), .groups = "drop")
```

```r
mig_rate_final <- mig_rate %>%
  group_by(country_name) %>%
  summarise(mig_rate = mean(mig_rate, na.rm = TRUE), .groups = "drop")

unemployment_final <- unemployment %>%
  group_by(country_name) %>%
  summarise(youth_unempl_rate = mean(youth_unempl_rate, na.rm = TRUE), .groups = "drop")

joined_df <- ed_exp_final %>%
  full_join(mig_rate_final, by = "country_name") %>%
  full_join(unemployment_final, by = "country_name")

dim(joined_df)
```

```
## [1] 227   4
```

c. Enriching Data with Income Classification

Obtain country income classification (low, lower-middle, upper-middle, high) from the World Bank and merge it with the dataset.

- Identify common variables between datasets. Can they be used for merging? Why or why not?

The common variable between joined_df and the World Bank dataset is the country name (Economy column). However we cannot use it directly for merging because the Economy column also contains region names (like "South Asia" or "World") which are not countries and we would therefore get unwanted rows. Also, the country names could be different depending on the source. Therefore we use ISO codes as the merge key instead, because these are standardized and uniquely identify each country. The World Bank file contains a column named Code with ISO codes, and the region names don't have a code so we remove them with filter(!is.na(iso_code)).

- Since ISO codes are standardized, download and use the CIA country data codes for merging. Make sure you are not losing any of the countries in your original data set when merging.

We use the CIA country data codes file to get the ISO codes for each country. They are then added to joined_df with a left join by country_name. Several countries had a dash - instead of a valid ISO code in the CIA file (for example France, India, Serbia and others), so their ISO codes are assigned manually using case_match. We then merge the World Bank data with another left join by iso_code.

```r
income_class <- read_excel("CLASS_2025_10_07.xlsx") %>%
  select(Code, "Income group") %>%
  rename(iso_code = Code, income_group = "Income group") %>%
  filter(!is.na(iso_code))

iso_country <- read_csv("Country Data Codes.csv") %>%
  select("Name", "Stanag") %>%
  rename(country_name = Name, iso_code = Stanag) %>%
  mutate(iso_code = case_match(country_name,
      "Kosovo"       ~ "XKX",
      "Saint Martin" ~ "MAF",
      .default = iso_code),
      country_name = countrycode(country_name, origin = 'country.name', destination = 'country.name', cu
```

```
## Rows: 278 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
```

```
## chr (6): Name, GENC, ISO 3166, Stanag, Internet, Comment
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `country_name = countrycode(...)`.
## Caused by warning:
## ! Some values were not matched unambiguously: Akrotiri, Ashmore and Cartier Islands, Baker Island, Cl
## To fix unmatched values, please use the `custom_match` argument. If you think the default matching ru
```

```r
iso_df <- joined_df %>%
  left_join(iso_country, by = "country_name") %>%
  mutate(iso_code = case_match(country_name,
    "Guernsey"               ~ "GGY",
    "Isle of Man"            ~ "IMN",
    "Jersey"                 ~ "JEY",
    "North Macedonia"        ~ "MKD",
    "France"                 ~ "FRA",
    "India"                  ~ "IND",
    "Serbia"                 ~ "SRB",
    "Myanmar (Burma)"        ~ "MMR",
    "Samoa"                  ~ "WSM",
    "South Sudan"            ~ "SSD",
    "Congo - Kinshasa"       ~ "COD",
    "British Virgin Islands" ~ "VGB",
    "Curaçao"                ~ "CUW",
    "Sint Maarten"           ~ "SXM",
    "St. Barthélemy"         ~ "BLM",
    .default = iso_code)) %>%
  left_join(income_class, by = "iso_code")

dim(iso_df)
```

```
## [1] 233   6
```

d. Adding Geographical Information

Introduce continent and subcontinent (or region) data for each country.

- Find and download an appropriate online resource.
- Merge this information into the dataset, naming the final dataset `df_vars`. Make sure you are not losing any of the countries in your original data set when merging.

We use a country-continent dataset from kaggle which contains ISO codes, continent and sub-region for each country. We merge it to iso_df with a left join by iso_code.

```r
#  https://www.kaggle.com/datasets/andradaolteanu/country-mapping-iso-continent-region/data
continent_df <- read_csv("continents2.csv") %>%
  select("alpha-3", "region", "sub-region")%>%
  rename("iso_code" = "alpha-3", "continent" = "region", "region" = "sub-region")
```

```
## Rows: 249 Columns: 11
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (7): name, alpha-2, alpha-3, iso_3166-2, region, sub-region, intermediat...
## dbl (4): country-code, region-code, sub-region-code, intermediate-region-code
```

```
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
df_vars <- iso_df %>%
  left_join(continent_df, by = "iso_code")
```

e. Data Tidiness and Summary Statistics

- Evaluate the tidiness of `df_vars` (observational units, variables, fixed vs. measured variables). Make adjustments to tidy the data, if necessary.

df_vars had one tidiness issue. It had a few duplicate rows caused by some countries appearing multiple times in the CIA codes file under different name variants, which after countrycode() standardisation mapped to the same country name. These were removed with distinct(). Also, West Bank and Gaza Strip have the same ISO code (PSE), but have different country names and slightly different values, so both are intentionally kept as distinct territories. After, we also moved fixed variables to the front of the dataset, followed by measured variables.

- Create a frequency table for the income status variable and briefly interpret the results.

When looking at the frequency table for income status, we can observe that the majority of countries are classified as High income (86 countries, 37.9%), followed by Upper middle income (54 countries, 23.8%), Lower middle income (51 countries, 22.5%) and Low income (25 countries, 11%). 11 countries have no income classification. This reflects the global distribution of countries, where high income countries make up the largest group, as there are many small countries that are wealthy.

- Analyze the distribution of income status across continents by computing absolute and relative frequencies. Comment on the findings.

The distribution of income status across continents varies quite heavily across continents and reveals some economic patterns. In Europe around 80% of the countries are classified as High income and the remaining 16% being Upper middle income (we have two NA's). The Americas are also relatively wealthy with 46% of countries being High income and 36% being Upper middle income. On the other hand, Africa is at the other extreme, with most countries being Lower middle income (41.8%) and Low income (38.2%), and only 1 High income country. Asia is more balanced with most countries being Lower middle income (36.5%), followed by High income (28.8%) and Upper middle income (26.9%). Finally, Oceania is mainly dominated by High income countries (42.9%), and with Upper and Lower middle income countries having the same relative frequency (23.8%), which shows the mix of wealthy nations like Australia and New Zealand next to small Pacific island states.

- Using the distribution of income status across continents, identify which countries are the only ones in their income group across the continent. Discuss briefly.

We can see that there are only 2 countries that are the only representatives of their income group within their continent. Seychelles is the only High income country in Africa, which stands out because Africa is mainly dominated by Low and Lower middle income countries. After, Kosovo appears as the only Upper middle income country with no assigned continent (NA), which is expected as Kosovo is not recognized universally and therefore has no match in the continent dataset.

```r
#tidying the data

df_vars <- df_vars %>%
  distinct(country_name, .keep_all = TRUE) %>%
  select(country_name, iso_code, income_group, continent, region,
         ed_expenditure, mig_rate, youth_unempl_rate)

glimpse(df_vars)
```

```
## Rows: 227
## Columns: 8
## $ country_name      <chr> "Afghanistan", "Albania", "Andorra", "Angola", "Arge~
## $ iso_code          <chr> "AFG", "ALB", "AND", "AGO", "ARG", "ARM", "ABW", "AU~
## $ income_group      <chr> "Low income", "Upper middle income", "High income", ~
## $ continent         <chr> "Asia", "Europe", "Europe", "Africa", "Americas", "A~
## $ region            <chr> "Southern Asia", "Southern Europe", "Southern Europe~
## $ ed_expenditure    <dbl> 4.1, 3.6, 3.2, 3.4, 5.5, 2.7, 5.5, 5.1, 5.4, 2.5, 2.~
## $ mig_rate          <dbl> -0.1, -3.3, 0.0, -0.2, -0.1, -5.5, 8.4, 8.1, 3.6, 0.~
## $ youth_unempl_rate <dbl> 17.6, 31.9, NA, 39.4, 23.7, 36.3, NA, 11.8, 9.4, 13.~
```

```r
df_vars %>%
  count(iso_code) %>%
  filter(n > 1) %>%
  print()
```

```
## # A tibble: 1 x 2
##   iso_code     n
##   <chr>    <int>
## 1 PSE          2
```

```r
# Frequency Table
income_freq <- df_vars %>%
  count(income_group) %>%
  mutate(percentage = n / sum(n) * 100)

print(income_freq)
```

```
## # A tibble: 5 x 3
##   income_group            n percentage
##   <chr>               <int>      <dbl>
## 1 High income            86      37.9
## 2 Low income             25      11.0
## 3 Lower middle income    51      22.5
## 4 Upper middle income    54      23.8
## 5 <NA>                   11       4.85
```

```r
# Relative Frequency Table
continent_income_dist <- df_vars %>%
  group_by(continent, income_group) %>%
  summarise(n = n(), .groups = "drop_last") %>%
  mutate(rel_freq = n / sum(n) * 100)

print(continent_income_dist)
```

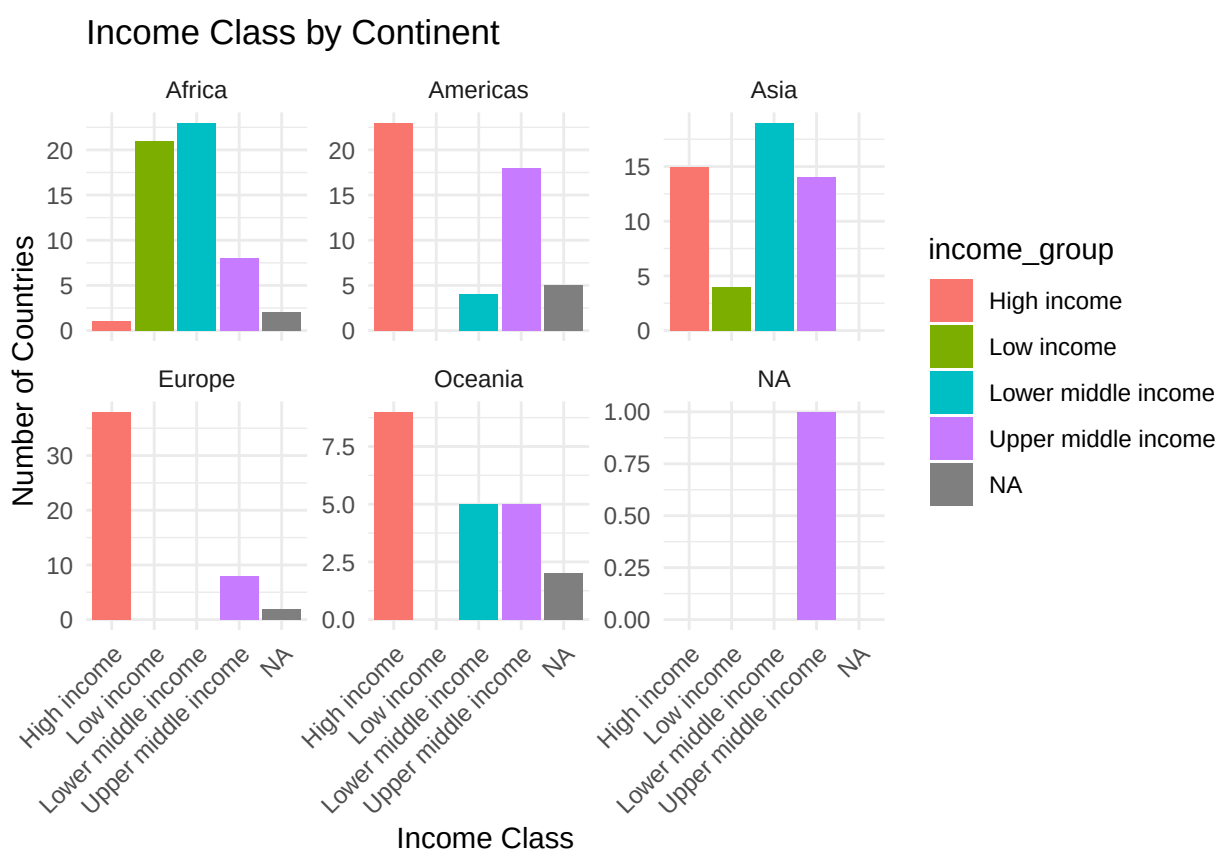
```
## # A tibble: 21 x 4
## # Groups:   continent [6]
##    continent income_group            n rel_freq
##    <chr>     <chr>               <int>    <dbl>
##  1 Africa    High income             1     1.82
##  2 Africa    Low income             21    38.2
##  3 Africa    Lower middle income    23    41.8
##  4 Africa    Upper middle income     8    14.5
##  5 Africa    <NA>                    2     3.64
##  6 Americas  High income            23    46
##  7 Americas  Lower middle income     4     8
```

```
##   8 Americas  Upper middle income    18    36
##   9 Americas  <NA>                    5    10
## 10 Asia       High income            15    28.8
## # i 11 more rows
```

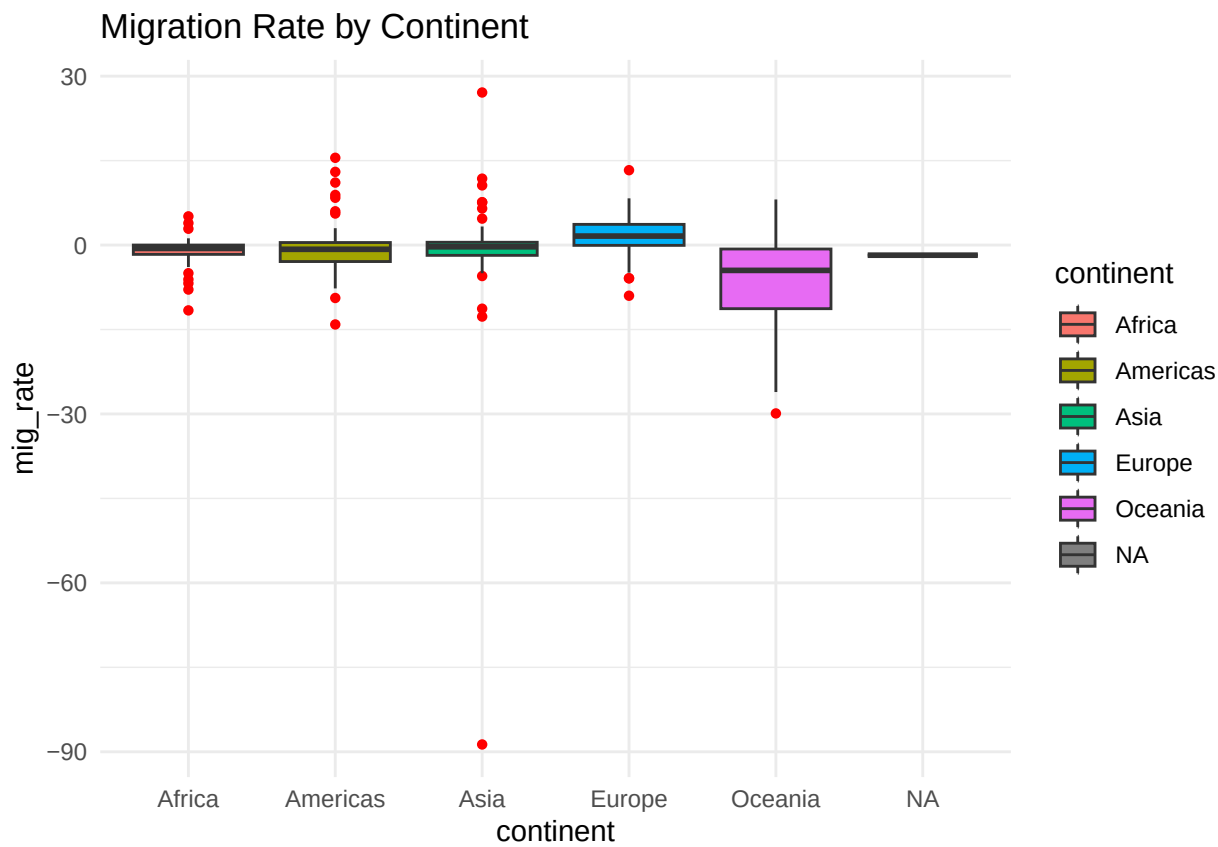
```
unique_income_countries <- df_vars %>%
  group_by(continent, income_group) %>%
  filter(n() == 1) %>%
  select(continent, country_name, income_group)

print(unique_income_countries)
```

```
## # A tibble: 2 x 3
## # Groups:   continent, income_group [2]
##   continent country_name income_group
##   <chr>     <chr>        <chr>
## 1 Africa    Seychelles   High income
## 2 <NA>      Kosovo       Upper middle income
```

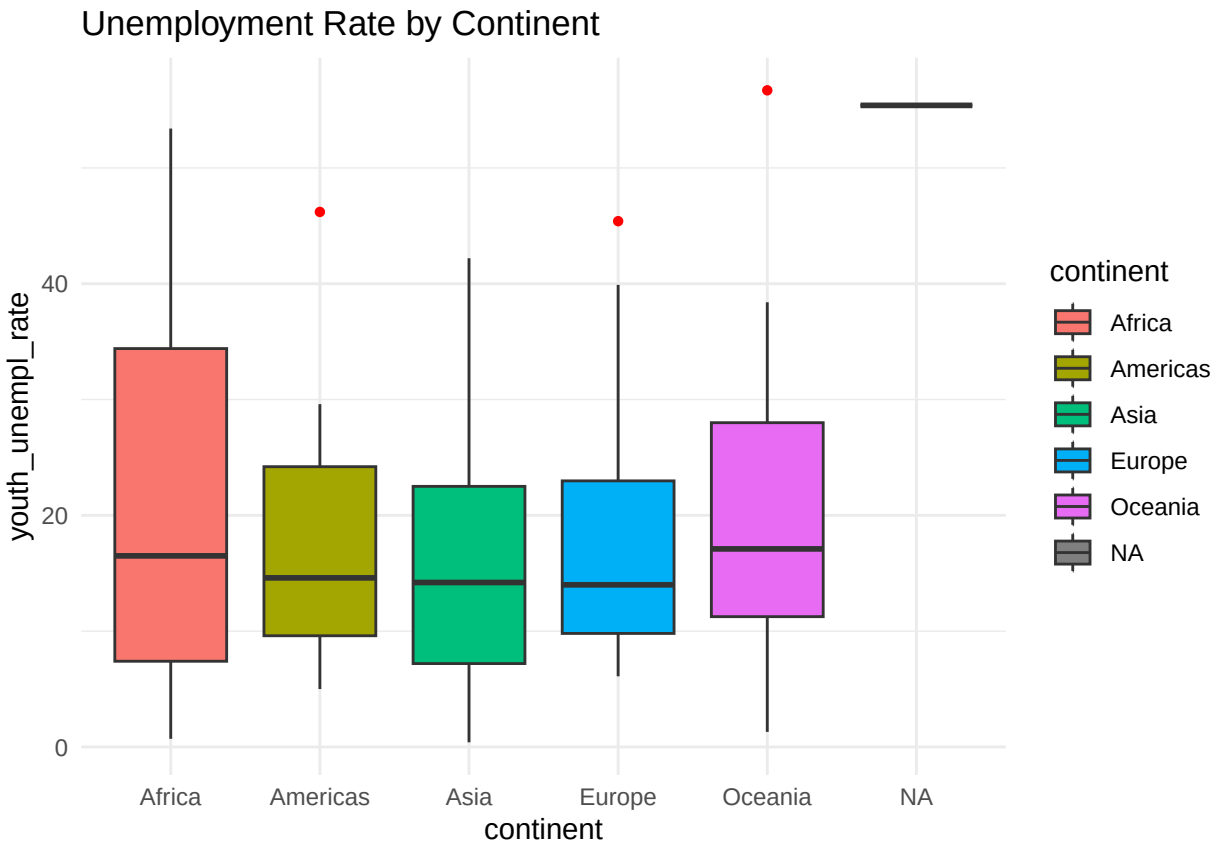
```
ggplot(df_vars, aes(x = income_group, fill = income_group)) +
  geom_bar() +
  facet_wrap(~ continent, scales = "free_y") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Income Class by Continent",
       x = "Income Class",
       y = "Number of Countries")
```



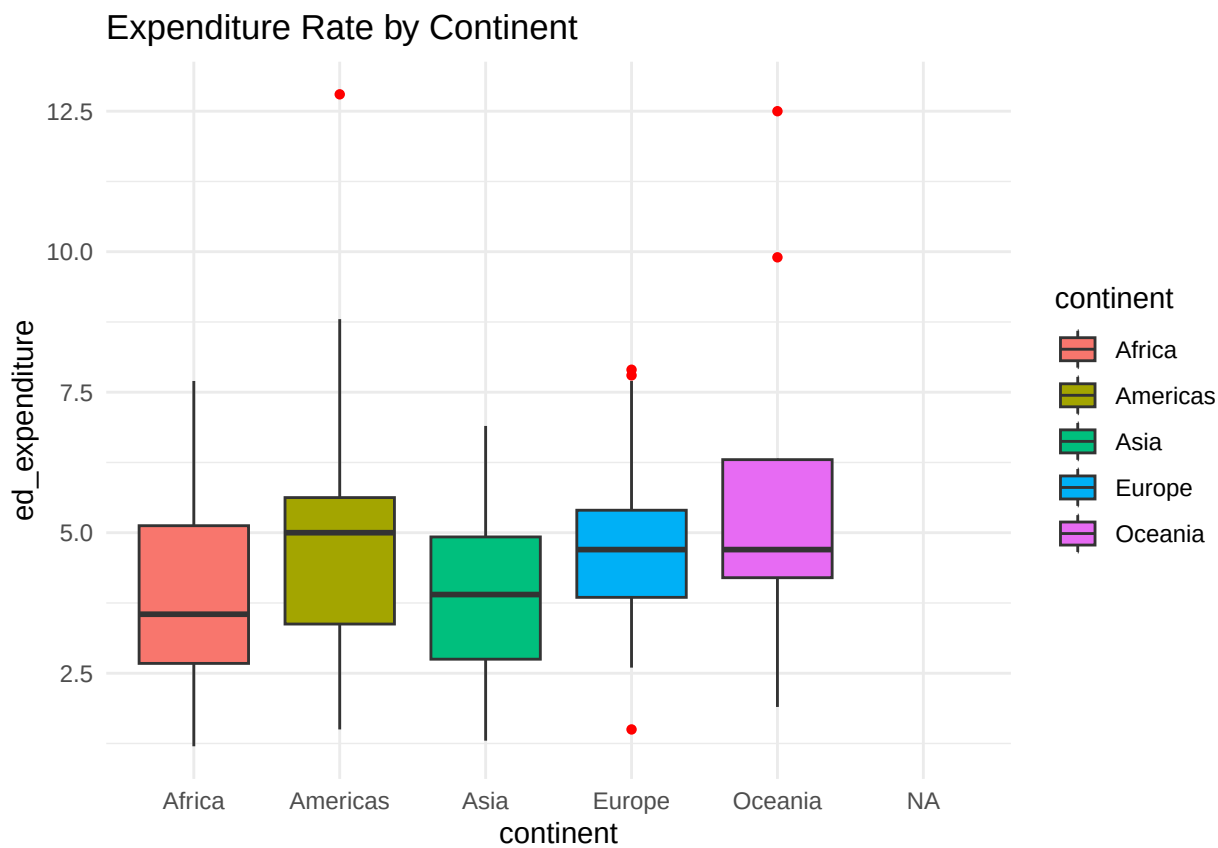
```r
ggplot(df_vars, aes(x = continent, y = mig_rate, fill = continent)) +
  geom_boxplot(outlier.color = "red", outlier.shape = 16) +
  theme_minimal() +
  labs(title = "Migration Rate by Continent")
```




```r
ggplot(df_vars, aes(x = continent, y = youth_unempl_rate, fill = continent)) +
  geom_boxplot(outlier.color = "red", outlier.shape = 16, na.rm = TRUE) +
  theme_minimal() +
  labs(title = "Unemployment Rate by Continent")
```




```r
ggplot(df_vars, aes(x = continent, y = ed_expenditure, fill = continent)) +
  geom_boxplot(outlier.color = "red", outlier.shape = 16, na.rm = TRUE) +
  theme_minimal() +
  labs(title = "Expenditure Rate by Continent")
```



f. Further Summary Statistics and Insights

- Create a table of average (mean and median) values for expenditure, youth unemployment rate and net migration rate separated into income status. Make sure that in the output, the ordering of the income classes is proper (i.e., L, LM, UM, H or the other way around). Briefly comment the results and any differences between the mean and median.

We can see that the mean education expenditure increases slightly with income level (from 3.5% to 4.6%). The mean and median are relatively close to each other, which suggests that the data is not strongly skewed. Net migration is only positive for High income countries, which shows that these countries are the main destination for migrants, with all the other groups having negative migration values. The difference between mean and median is larger for Lower middle income (-3.87 vs -1.30), which can suggest that there are some countries with very large negative migration rates that are pulling the mean down. For youth unemployment, Upper middle income countries have the highest mean (22.8%), which may be a little surprising.

```
df_summary <- df_vars %>%
  group_by(income_group) %>%
  summarize(exp_mean = mean(ed_expenditure, na.rm = TRUE),
            exp_median = median(ed_expenditure, na.rm = TRUE),
            mig_mean = mean(mig_rate, na.rm = TRUE),
            mig_median = median(mig_rate, na.rm = TRUE),
            unempl_mean = mean(youth_unempl_rate, na.rm = TRUE),
            unempl_median = median(youth_unempl_rate, na.rm = TRUE)) %>%
  mutate(income_group = factor(income_group,
                               levels = c("Low income",
                                          "Lower middle income",
                                          "Upper middle income",
```

```r
                                        "High income"))) %>%
  arrange(income_group)

print(df_summary)
```

```
## # A tibble: 5 x 7
##   income_group exp_mean exp_median mig_mean mig_median unempl_mean unempl_median
##   <fct>           <dbl>      <dbl>    <dbl>      <dbl>       <dbl>         <dbl>
## 1 Low income       3.53        3.1   -0.552      -0.7         14.9          11.2
## 2 Lower middl~     4.47        4.1   -3.87       -1.3         17.0          14
## 3 Upper middl~     4.51        4.1   -2.34       -0.85        22.8          20.2
## 4 High income      4.6         4.7    1.45        1.65        16.5          13.2
## 5 <NA>             6.07        4.7   -3.02       -0.2         19.9          19.9
```

- Look at the standard deviation and the interquartile range of the variables per income status instead of the location statistics above. Do you gain additional insights? Briefly comment the results.

For education expenditure, Low and Lower middle income countries have a higher SD and IQR than High and Upper middle income countries, which shows more variability in how these countries invest in education. For the net migration rate, we can see that Lower middle income countries have a much larger SD (12.7) compared to the IQR (2.6), which shows that there are a few outliers that make the SD much larger. High income countries have the largest migration IQR (5.1), which shows that there is quite a mix between countries that are destinations for migration and countries that experience emigration. For youth unemployment, Upper middle income countries have the largest IQR, which suggests quite big differences in this group of countries. Overall, when SD is much larger than IQR it is a sign of outliers.

```r
df_sd <- df_vars %>%
  group_by(income_group) %>%
  summarize(exp_sd = sd(ed_expenditure, na.rm = TRUE),
            exp_iqr = IQR(ed_expenditure, na.rm = TRUE),
            mig_sd = sd(mig_rate, na.rm = TRUE),
            mig_iqr = IQR(mig_rate, na.rm = TRUE),
            unempl_sd= sd(youth_unempl_rate, na.rm = TRUE),
            unempl_iqr = IQR(youth_unempl_rate, na.rm = TRUE)
            )

print(df_sd)
```

```
## # A tibble: 5 x 7
##   income_group        exp_sd exp_iqr mig_sd mig_iqr unempl_sd unempl_iqr
##   <chr>                <dbl>   <dbl>  <dbl>   <dbl>     <dbl>      <dbl>
## 1 High income           1.49    1.8    6.24    5.05     10.3        14.0
## 2 Low income            1.73    2.65   6.28    2.9      12.8        15.1
## 3 Lower middle income   2.20    2.4   12.7     2.6      12.7        15.4
## 4 Upper middle income   1.78    1.65   3.87    4.12     13.5        17.5
## 5 <NA>                  2.37    2.05  10.1     4.8       7.50        5.3
```

- Extend the analysis of the statistics median and IQR to **each income status and continent combination**. Play around with displaying the resulting table. Use `pivot_longer()` and/or `pivot_wider()` to generate different outputs. Discuss the results as well as the readability of the different tables.

When looking at the results, we can see that high income countries in Europe show a relatively low IQR for education expenditure (1.45) and migration (4.4), which suggests that there are consistent patterns across European high income countries. In Oceania, high income countries stand out with a very large migration IQR (15.1), which shows the mix between major destination countries like Australia and small island countries with negative migration. Also, upper middle income countries show interesting patterns across continents.

Africa stands out with the highest unemployment median (36%) and also a large unemployment IQR (12.25). Europe countries have a really large unemployment median (29.55%) with a large unemployment IQR (16.3), which shows the mix of countries with high unemployment and wealthier countries with lower unemployment. Oceania has a very low education expenditure IQR (0.15) suggesting similar spending patterns, but a large unemployment IQR (9.125).

The base table (21x8) is the most readable, where we have one row per income-continent combination and the different statistics as columns. The long format (126x4) has one row per statistic, which makes it too long to read. The wide format (5x37) has only 5 rows but 37 columns, which makes it also hard to read.

```
df_median_iqr_continent <- df_vars %>%
  group_by(income_group, continent) %>%
  summarize(exp_median = median(ed_expenditure, na.rm = TRUE),
            exp_iqr = IQR(ed_expenditure, na.rm = TRUE),
            mig_median = median(mig_rate, na.rm = TRUE),
            mig_iqr = IQR(mig_rate, na.rm = TRUE),
            unempl_median = median(youth_unempl_rate, na.rm = TRUE),
            unempl_iqr = IQR(youth_unempl_rate, na.rm = TRUE)
            )
```

```
## `summarise()` has grouped output by 'income_group'. You can override using the
## `.groups` argument.
```

```
print(df_median_iqr_continent)
```

```
## # A tibble: 21 x 8
## # Groups:   income_group [5]
##    income_group    continent exp_median exp_iqr mig_median mig_iqr unempl_median
##    <chr>           <chr>          <dbl>   <dbl>      <dbl>   <dbl>         <dbl>
##  1 High income     Africa          4.4    0           1       0             11.6
##  2 High income     Americas        5      2.4         0.3     5.4           21.0
##  3 High income     Asia            3.8    2.3         2.3     5.8            8.9
##  4 High income     Europe          4.7    1.45        2.1     4.4           12.4
##  5 High income     Oceania         5.7    0.600      -0.7    15.1           26.6
##  6 Low income      Africa          2.95   2.82       -0.9     3.2            8.7
##  7 Low income      Asia            4.1    0          -0.05    6.9           24.5
##  8 Lower middle i~ Africa          3.55   2.05       -0.4     1.65          16.5
##  9 Lower middle i~ Americas        5.25   2.4        -1.65    0.9            8.5
## 10 Lower middle i~ Asia            4      2.7        -1.8     3.1           13.7
## # i 11 more rows
## # i 1 more variable: unempl_iqr <dbl>
```

```
df_median_iqr_long <- df_median_iqr_continent %>%
  pivot_longer(
    cols = exp_median:unempl_iqr,
    names_to = "statistic",
    values_to = "value"
  )
print(df_median_iqr_long)
```

```
## # A tibble: 126 x 4
## # Groups:   income_group [5]
##    income_group continent statistic     value
##    <chr>        <chr>     <chr>         <dbl>
##  1 High income  Africa    exp_median      4.4
##  2 High income  Africa    exp_iqr         0
##  3 High income  Africa    mig_median      1
```

```
##  4 High income  Africa    mig_iqr         0
##  5 High income  Africa    unempl_median  11.6
##  6 High income  Africa    unempl_iqr      0
##  7 High income  Americas  exp_median      5
##  8 High income  Americas  exp_iqr         2.4
##  9 High income  Americas  mig_median      0.3
## 10 High income  Americas  mig_iqr         5.4
## # i 116 more rows
```

```
df_median_iqr_wide <- df_median_iqr_continent %>%
  pivot_wider(
    names_from = continent,
    values_from = exp_median:unempl_iqr
  )
print(df_median_iqr_wide)
```

```
## # A tibble: 5 x 37
## # Groups:   income_group [5]
##   income_group        exp_median_Africa exp_median_Americas exp_median_Asia
##   <chr>                           <dbl>               <dbl>           <dbl>
## 1 High income                      4.4                 5                3.8
## 2 Low income                       2.95               NA                4.1
## 3 Lower middle income              3.55                5.25             4
## 4 Upper middle income              4.95                4.75             3.8
## 5 <NA>                             4.7                 8.8             NA
## # i 33 more variables: exp_median_Europe <dbl>, exp_median_Oceania <dbl>,
## #   exp_median_NA <dbl>, exp_iqr_Africa <dbl>, exp_iqr_Americas <dbl>,
## #   exp_iqr_Asia <dbl>, exp_iqr_Europe <dbl>, exp_iqr_Oceania <dbl>,
## #   exp_iqr_NA <dbl>, mig_median_Africa <dbl>, mig_median_Americas <dbl>,
## #   mig_median_Asia <dbl>, mig_median_Europe <dbl>, mig_median_Oceania <dbl>,
## #   mig_median_NA <dbl>, mig_iqr_Africa <dbl>, mig_iqr_Americas <dbl>,
## #   mig_iqr_Asia <dbl>, mig_iqr_Europe <dbl>, mig_iqr_Oceania <dbl>, ...
```

- Identify countries performing well in terms of both **youth unemployment** and **net migration rate** (top 25% in net migration and bottom 25% in youth unemployment within their continent).

Countries that perform well in both youth unemployment and net migration rate are for example Norway and Switzerland in Europe, which makes sense given that they both have strong economies. Also countries like Qatar, Bahrain and UAE are among those performing well, which is to be expected. Furthermore, we removed Kosovo from the table below, which would appear because it isn't assigned a continent (NA), but is massively under-performing for Europe.

```
df_performing <- df_vars %>%
  filter(!is.na(continent)) %>%
  group_by(continent) %>%
  mutate( mig_quant = quantile(mig_rate, probs = 0.75, na.rm = TRUE),
          unempl_quant= quantile(youth_unempl_rate, probs=0.25, na.rm = TRUE)) %>%
  filter(mig_rate >= mig_quant & youth_unempl_rate <= unempl_quant) %>%
  ungroup() %>%
  select(continent, country_name, youth_unempl_rate, unempl_q = unempl_quant, mig_rate, mig_q = mig_quan
  arrange(continent)

print(df_performing)
```

```
## # A tibble: 16 x 6
##    continent country_name         youth_unempl_rate unempl_q mig_rate mig_q
```

```
##    <chr>     <chr>                            <dbl>    <dbl>    <dbl> <dbl>
##  1 Africa    Benin                              5.6      7.4      0.3  0
##  2 Africa    Côte d'Ivoire                      5.5      7.4      1.2  0
##  3 Africa    Guinea                             1        7.4      0    0
##  4 Africa    Madagascar                         1        7.4      0    0
##  5 Africa    Togo                               3.9      7.4      0    0
##  6 Americas  United States                      8.6      9.6      3    0.45
##  7 Asia      Bahrain                            5.3      7.2     10.6  0.5
##  8 Asia      Israel                             7.2      7.2      2.1  0.5
##  9 Asia      Macao SAR China                    5.3      7.2      3.3  0.5
## 10 Asia      Qatar                              0.4      7.2      6.5  0.5
## 11 Asia      United Arab Emirates               6.9      7.2      7.6  0.5
## 12 Europe    Malta                              9.1      9.8      6.6  3.68
## 13 Europe    Norway                             9.7      9.8      4    3.68
## 14 Europe    Switzerland                        7.9      9.8      4.6  3.68
## 15 Oceania   Papua New Guinea                   3.6     11.2      0   -0.7
## 16 Oceania   Palau                              5.6     11.2      0.9 -0.7
```

g. Conditional Probabilities

Estimate the following based on the observed frequencies in the data:

- What is the (posterior or conditional) probability that a European country belongs to the high income group? What is the prior probability that a country belongs to the high income group?

The conditional probability that a European country belongs to the high income group is 0.83, which makes intuitive sense since Europe is mainly composed of high income countries. The prior probability that any country in the dataset belongs to the high income group is 0.4, which is as expected much lower than the probability of a European country belonging to the high income group.

```r
prob_europe <- df_vars %>%
  filter(continent == "Europe") %>%
  summarize(conditional_prob = mean(income_group == "High income", na.rm = TRUE))
print(prob_europe)
```

```
## # A tibble: 1 x 1
##   conditional_prob
##              <dbl>
## 1            0.826
```

```r
prob_overall<- df_vars %>%
  summarize(prior_prob = mean(income_group == "High income", na.rm = TRUE))
print(prob_overall)
```

```
## # A tibble: 1 x 1
##   prior_prob
##        <dbl>
## 1      0.398
```

- Given a country has high youth unemployment (above %25), what is the probability that it also has negative net migration?

Given that a country has a youth unemployment rate above 25%, the probability that it also has a negative net migration is 0.61. This makes sense as countries with high unemployment tend to have a worse economy, which can drive younger people to emigrate, which results in negative net migration.

```r
prob_neg_mig <- df_vars %>%
  filter(youth_unempl_rate > 25) %>%
```

```r
  summarize(prob = mean(mig_rate < 0, na.rm = TRUE))

print(prob_neg_mig)
```

```
## # A tibble: 1 x 1
##    prob
##   <dbl>
## 1 0.612
```

h. Simpson's Paradox Analysis

Investigate whether an overall trend in youth unemployment rate in the high and low income groups reverses when analyzed at the continent level. E.g., does the youth unemployment rate appear lower in low-income countries overall, but higher when controlling for continent? Explain the results and possible reasons behind this paradox.

```r
hl_df <- df_vars %>%
  filter(income_group %in% c("High income", "Low income")) %>%
  drop_na(youth_unempl_rate, income_group, continent)

overall_unemp <- hl_df %>%
  group_by(income_group) %>%
  summarise(
    mean_youth_unempl = mean(youth_unempl_rate, na.rm = TRUE),
    median_youth_unempl = median(youth_unempl_rate, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )

overall_unemp
```

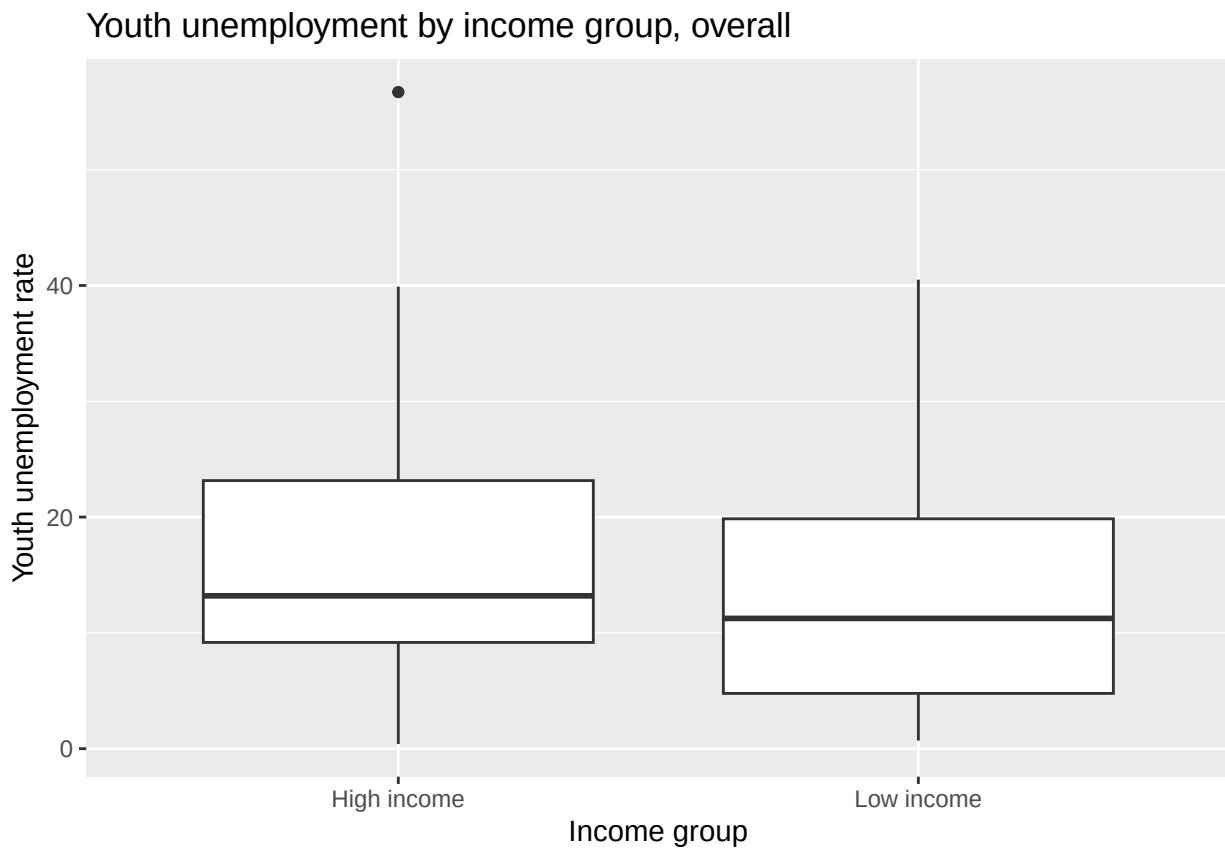
```
## # A tibble: 2 x 4
##   income_group mean_youth_unempl median_youth_unempl     n
##   <chr>                    <dbl>               <dbl> <int>
## 1 High income               16.5                13.2    70
## 2 Low income                14.9                11.2    18
```

```r
continent_unemp <- hl_df %>%
  group_by(continent, income_group) %>%
  summarise(
    mean_youth_unempl = mean(youth_unempl_rate, na.rm = TRUE),
    median_youth_unempl = median(youth_unempl_rate, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )

continent_unemp
```

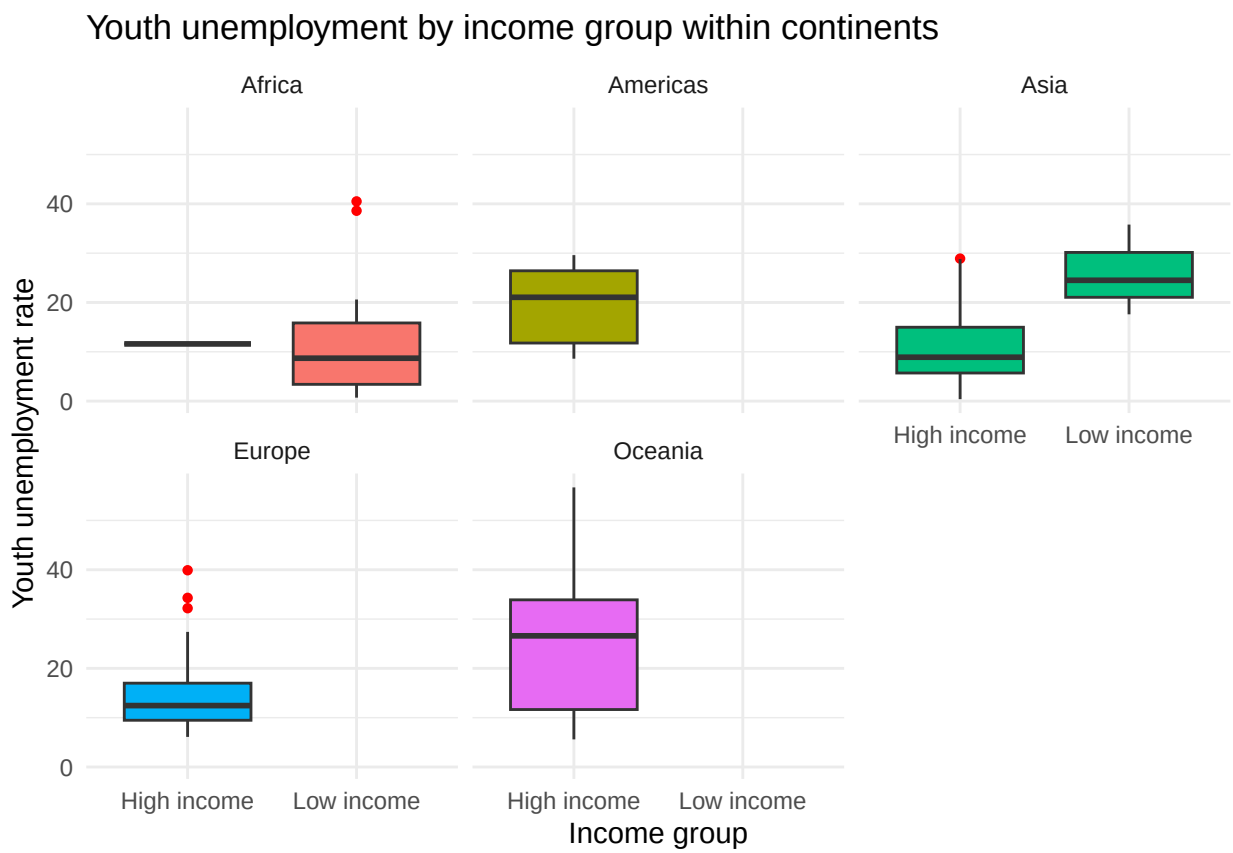
```
## # A tibble: 7 x 5
##   continent income_group mean_youth_unempl median_youth_unempl     n
##   <chr>     <chr>                    <dbl>               <dbl> <int>
## 1 Africa    High income               11.6                11.6     1
## 2 Africa    Low income                12.6                 8.7    15
## 3 Americas  High income               19.9                21.0    14
## 4 Asia      High income               11.7                 8.9    14
```

```
## 5 Asia      Low income                26.0                24.5     3
## 6 Europe    High income               15.4                12.4    34
## 7 Oceania   High income               25.7                26.6     7
```

```
ggplot(hl_df, aes(x = income_group, y = youth_unempl_rate)) +
  geom_boxplot() +
  labs(
    x = "Income group",
    y = "Youth unemployment rate",
    title = "Youth unemployment by income group, overall"
  )
```




```
ggplot(hl_df, aes(x = income_group, y = youth_unempl_rate, fill = continent)) +
  geom_boxplot(outlier.color = "red", outlier.shape = 16) +
  facet_wrap(~ continent) +
  theme_minimal() +
  theme(legend.position = "none") +
  labs(
    x = "Income group",
    y = "Youth unemployment rate",
    title = "Youth unemployment by income group within continents"
  )
```



For this analysis, we only compare high-income and low-income countries. We compare the countries overall and within each continent using the mean youth unemployment rate. Within continents, low-income countries appear to have higher youth unemployment, but overall this pattern reverses because the continent-level means and group sizes differ (Simpson's paradox). However, we only have data of both income groups for 2 out of 5 continents, so further analysis would be required.

The plot does not clearly show the reversal for Africa, because the median, which is emphasized by the plot, does not reverse, while the mean does. This happens because Africa has some outlier low-income countries with very high youth unemployment rates.

i. Data Export

Export the final tidy dataset from e. as a **CSV** with: `;` as a separator; `.` representing missing values; no row names included. Upload the `.csv` to TUWEL, together with the submission.

```
write.table(
  df_vars,
  "processed_data.csv",
  sep = ";",
  na = ".",
  row.names = FALSE,
  col.names = TRUE
)
```